

ReconForge Support Matrix

Version 1.0 — Last updated: 2026-03-21

Python Versions

Version	Status	Notes
3.10	✓ Supported (minimum)	<code>match</code> statements not used; <code>dataclass</code> features require 3.10+
3.11	✓ Tested	Primary development and CI target
3.12	✓ Supported	Compatible; no deprecated API usage
3.13+	⚠ Untested	Expected to work; not yet validated
3.9 and below	✗ Unsupported	Uses <code>list[str]</code> type hints (PEP 585), <code>dataclass(slots=True)</code> , <code> </code> union syntax

Operating Systems / Platforms

Platform	Status	Notes
Kali Linux (latest)	✓ Tested	Recommended. Most external tools pre-installed.
Parrot OS	✓ Supported	Similar tool availability to Kali.
Ubuntu 22.04+	✓ Supported	Tools available via <code>apt</code> / <code>pip</code> / <code>go install</code> .
Debian 12+	✓ Supported	Same package availability as Ubuntu.
Arch Linux	⚠ Untested	Tools available via <code>pacman</code> / AUR; expected to work.
macOS	⚠ Partial	Python core works. Some tools (<code>nmap</code> , <code>enum4linux</code>) available via Homebrew. <code>hydra</code> , <code>smbclient</code> may require manual setup. Root-requiring scans (<code>-sS</code> , <code>-sU</code>) need <code>sudo</code> .
Windows / WSL2	⚠ Partial	WSL2 with Kali/Ubuntu image works. Native Windows unsupported — <code>subprocess.run</code> assumes POSIX tool paths.
Docker	✓ Supported	Run in any Linux-based container with tools installed. No official Dockerfile yet.

Python Dependencies

Package	Required	Version	Purpose
<code>pyyaml</code>	✓ Yes	≥ 6.0	YAML config loading (<code>config_loader.py</code>)
<code>cryptography</code>	✗ Optional	≥ 41.0	Fernet encryption for loot vault (<code>--encrypt-loot</code>)
<code>pytest</code>	✗ Dev only	≥ 7.0	Test runner
<code>pytest-cov</code>	✗ Dev only	≥ 4.0	Coverage reporting

No other Python packages are required. The framework is intentionally dependency-light.

External Tools by Module

Missing tools are detected at runtime and their features are gracefully skipped. No tool absence causes a crash.

Network Module

Tool	Required	Install	Purpose
<code>nmap</code>	✓	<code>sudo apt install nmap</code>	Host discovery, port scanning, NSE scripts
<code>enum4linux</code>	✗	<code>sudo apt install enum4linux</code>	SMB/NetBIOS enumeration
<code>smbclient</code>	✗	<code>sudo apt install smbclient</code>	SMB share access
<code>ldapsearch</code>	✗	<code>sudo apt install ldap-utils</code>	LDAP queries
<code>hydra</code>	✗	<code>sudo apt install hydra</code>	Brute-force (opt-in only, requires <code>--brute-force</code> flag)

Web Module

Tool	Required	Install	Purpose
<code>gobuster</code>	✗	<code>go install git-hub.com/0J/gobuster/v3@latest</code>	Directory/file brute-forcing
<code>ffuf</code>	✗	<code>go install git-hub.com/ffuf/ffuf/v2@latest</code>	Content discovery, fuzzing
<code>nikto</code>	✗	<code>sudo apt install nikto</code>	Web vulnerability scanning
<code>nuclei</code>	✗	<code>go install git-hub.com/projectdiscovery/nuclei/v3@latest</code>	Template-based vuln scanning
<code>whatweb</code>	✗	<code>sudo apt install whatweb</code>	Technology fingerprinting
<code>wafw00f</code>	✗	<code>pip install wafw00f</code>	WAF detection
<code>wpscan</code>	✗	<code>gem install wpscan</code>	WordPress enumeration
<code>sqlmap</code>	✗	<code>sudo apt install sqlmap</code>	SQL injection detection (opt-in)
<code>curl</code>	✗	Pre-installed on most systems	HTTP probing

API Module

Tool	Required	Install	Purpose
ffuf	✗	<code>go install git-hub.com/ffuf/ffuf/v2@latest</code>	API endpoint fuzzing
httpx	✗	<code>go install git-hub.com/projectdiscovery/httpx/cmd/httpx@latest</code>	HTTP probing and tech detection
arjun	✗	<code>pip install arjun</code>	Hidden parameter discovery
nuclei	✗	<code>go install git-hub.com/projectdiscovery/nuclei/v3@latest</code>	API-specific vulnerability templates

Surface Module

Tool	Required	Install	Purpose
nmap	✓	<code>sudo apt install nmap</code>	Stealth port discovery

AD Module

Tool	Required	Install	Purpose
<code>nmap</code>	✓	<code>sudo apt install nmap</code>	AD service detection
<code>enum4linux-ng</code>	✗	<code>pip install enum4linux-ng</code>	Modern SMB/RPC enumeration
<code>impacket</code>	✗	<code>pip install impacket</code>	GetADUsers, GetN-PUUsers, lookupsid, rpcdump
<code>ldapsearch</code>	✗	<code>sudo apt install ldap-utils</code>	LDAP queries
<code>smbclient</code>	✗	<code>sudo apt install smbclient</code>	Share enumeration
<code>bloodhound-python</code>	✗	<code>pip install bloodhound</code>	BloodHound data collection
<code>netexec</code>	✗	<code>pip install netexec</code>	Multi-protocol enumeration

Root / Privileged Operations

Some operations require root or `CAP_NET_RAW` :

Operation	Why	Workaround
<code>nmap -sS</code> (SYN scan)	Raw socket access	Use <code>-sT</code> (connect scan) as non-root
<code>nmap -sU</code> (UDP scan)	Raw socket access	None — requires root
<code>nmap -O</code> (OS detection)	Raw socket access	Skip OS detection as non-root

ReconForge does not escalate privileges itself. Run with `sudo` when root-requiring profiles are needed.

Known Unsupported Environments

Environment	Reason
Python < 3.10	Modern type hint syntax, dataclass features
Native Windows (no WSL)	POSIX subprocess assumptions, tool path resolution
Alpine Linux (musl)	Some pip packages (<code>cryptography</code>) require build toolchain; fixable but untested
Restricted shells / containers without <code>subprocess</code>	Framework relies entirely on <code>subprocess.run</code> for tool execution
Air-gapped without pre-installed tools	No auto-download of external binaries; tools must be pre-installed